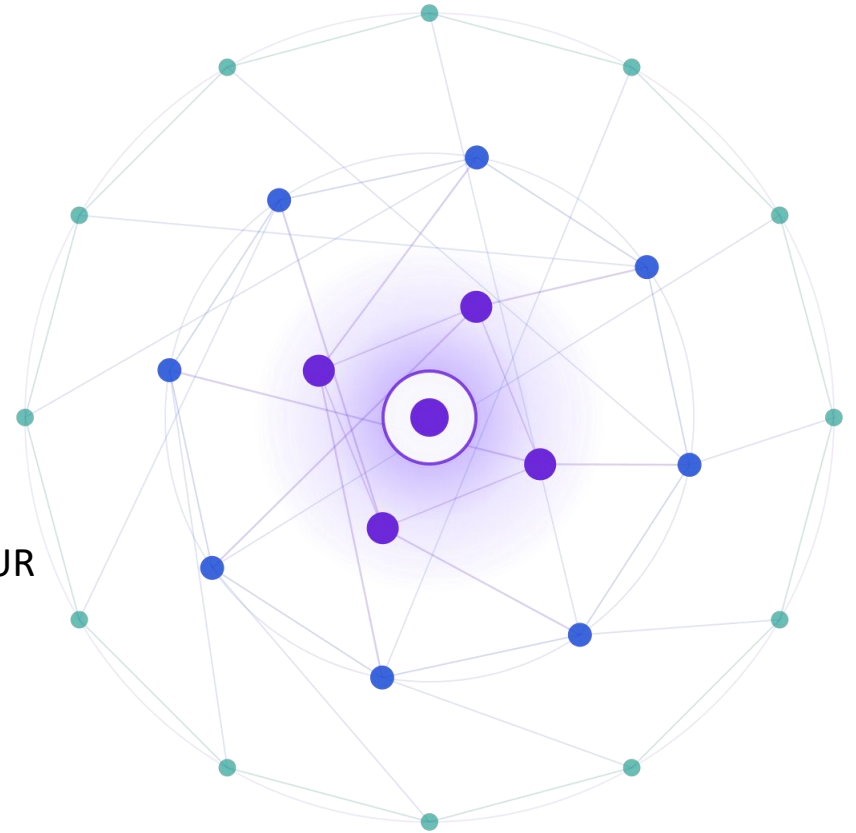


SAMSUNG PRISM

Generative AI Hackathon

3rd Edition 2026 – 27

- Theme ID - Theme 01 - Agentic Code Intelligence
- Team Name – SRMIST_JARVIS
- College Name – SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, KATTANKULATHUR
- Member Name & Email 1- Oviya B , ob1710@srmist.edu.in
- Member Name & Email 2- Padmanathan C , pc3604@srmist.edu.in
- Member Name & Email 3- Abinav K A , ak9500@srmist.edu.in
- Member Name & Email 4- Arhan Ashraf , aa7777@srmist.edu.in
- Submission Github link - https://github.com/abhinavakku513-collab/PRISM_GENAI_HACKATHON_Y2026



Theme

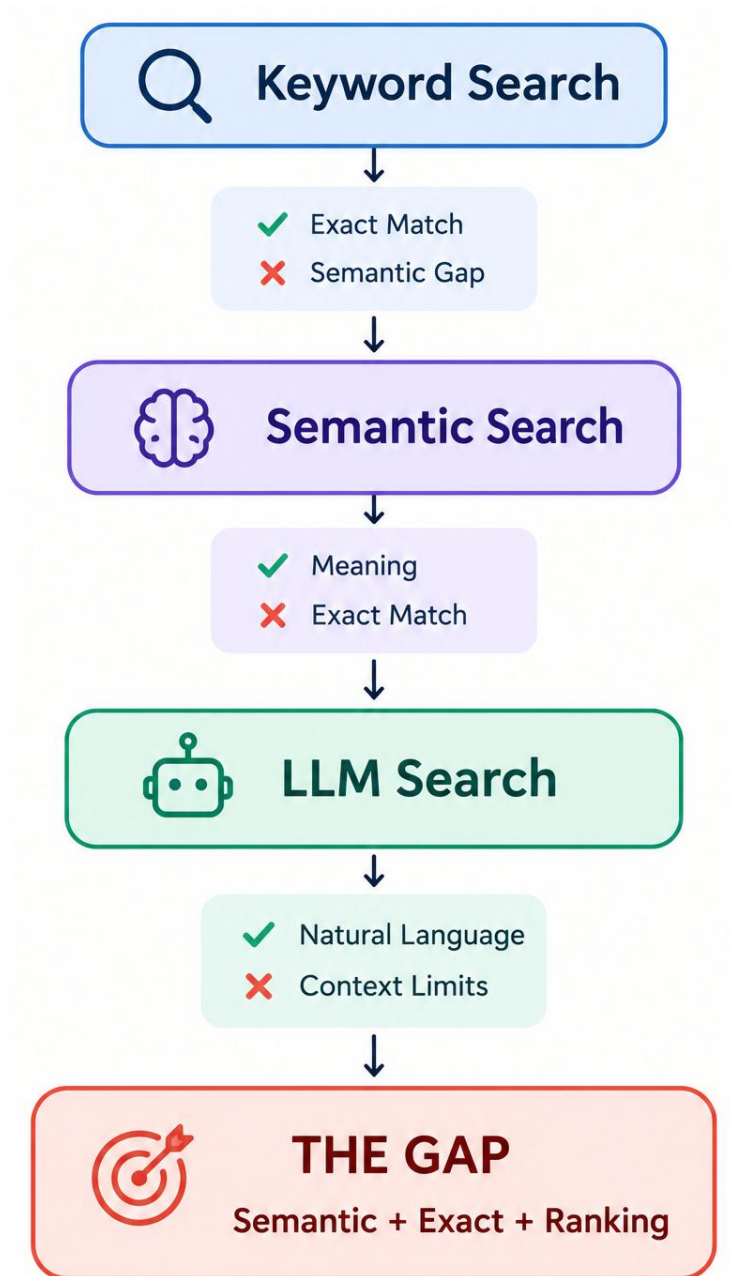
Theme 01 — Agentic Code Intelligence

- Modern software projects contain thousands of files, functions, and tools, making it difficult to quickly find the required code.
- Natural-language queries often do not directly match the terminology used in the codebase, making search less effective.
- Developers need an intelligent way to understand their query and retrieve the most relevant code with its exact file and line location.



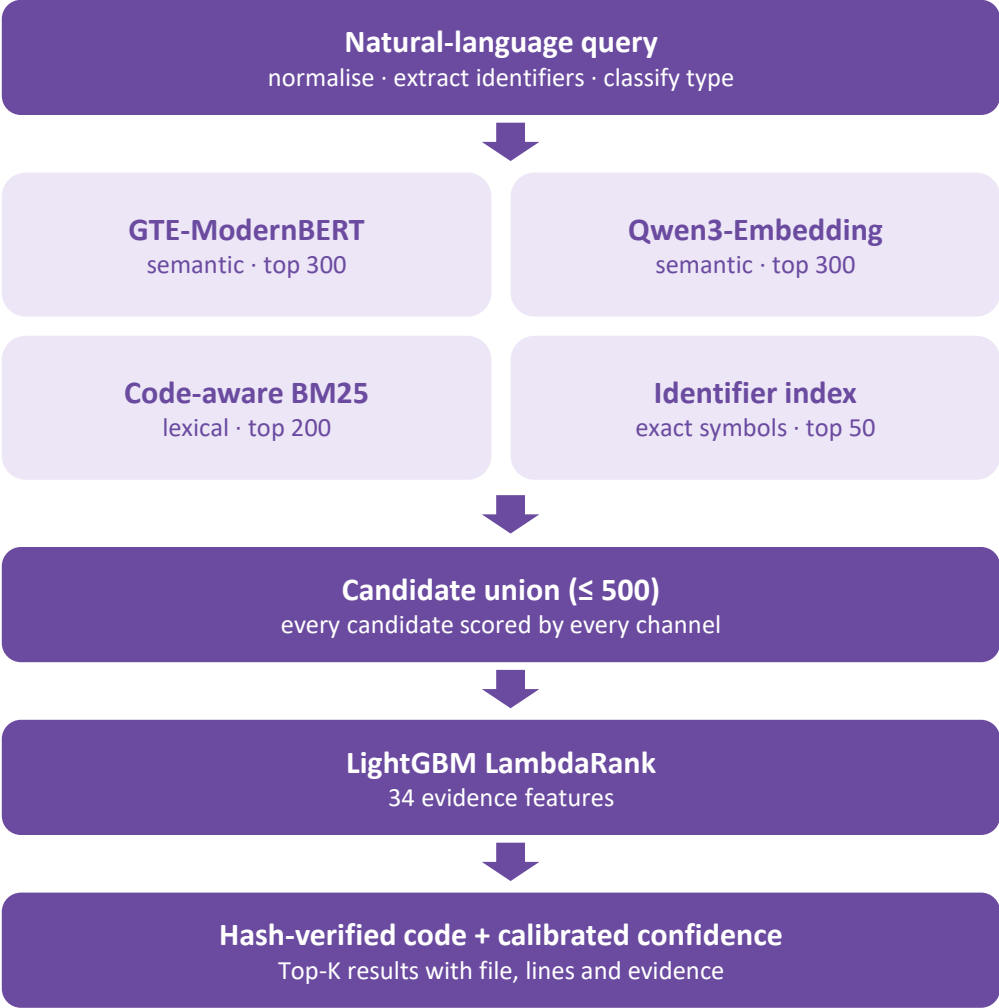
Existing Solutions & Gaps

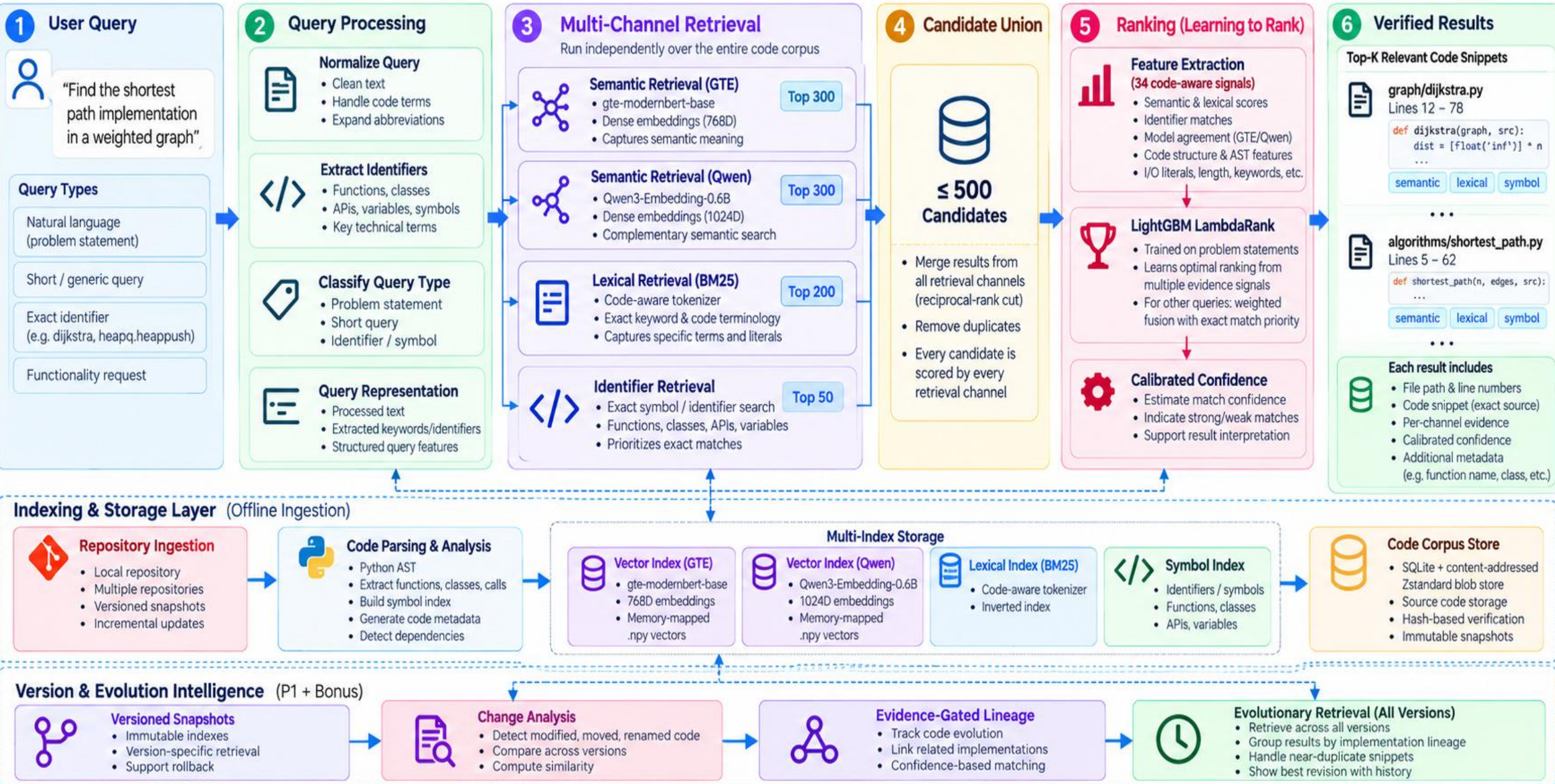
- Keyword search works well for exact terms but misses differently named code
- Semantic search understands meaning but can miss exact identifiers and symbols
- LLM-based search handles natural-language queries but struggles with large codebases
- Existing approaches often produce irrelevant results when multiple code sections are similar
- A unified approach is needed to combine **semantic understanding, exact matching, and intelligent ranking**



Our Solutions & Architecture Diagram

Architecture Component	Technology / Innovation	Purpose
Query Processing	Normalization + Identifier Extraction	Understands natural-language and code-specific queries
Query Routing	Query Type Classification	Identifies statement-like and short/generic queries
Semantic Retrieval	GTE-ModernBERT + Qwen3-Embedding-0.6B	Two dense encoders find semantically relevant code
Lexical Retrieval	BM25	Captures exact keyword/code terminology matches
Identifier Retrieval	Exact Symbol / Identifier Search	Prioritizes exact functions, classes and identifiers
Candidate Union	Multi-channel Retrieval	Combines candidates from all retrieval channels
Re-ranking	34 Ranking Signals + LightGBM LambdaRank	Selects the most relevant code
Code Verification	Hash-based Code Store	Re-reads and verifies the retrieved code
Confidence Layer	Calibrated Confidence	Indicates strong/weak matches
Version Intelligence	Immutable Snapshots + Incremental Updates	Supports version-specific retrieval and rollback
Code Evolution	Evidence-Gated Lineage	Tracks modified, moved, renamed and replaced code





ACIS

Hybrid code retrieval · learned ranking · offline

Search

APPS corpus

Versions

P1 · one pinned version

Evolution

Bonus · every version

Evaluation

P0 · recorded benchmark

Architecture

models · channels · ranker

Describe a problem. Get the code that solves it.

Every query runs through several independent retrieval channels, a candidate union and a learned ranker. Every result is stored code. re-read by its content hash: nothing is generated.

87.07

NDCG@10

84.72

MRR@10

94.34

HIT@10

98.76

RECALL@100

+16.04_{pt}

NDCG@10 VS DENSE ONLY

APPS dev split · 5,000 queries · out of fold · ledger dev:439f8a9f83a6 · chain and artifact verified

Official result, AppoRetrieval TEST split (MTEB): NDCG@10 78.28 · MRR@10 74.39 · ledger rc-bd2285393a88 — the files above are DEV numbers

Place n queens on an n by n chessboard so that no two queens attack each other. Return the number of distinct solutions.

ACIS full pipeline

Dense only

BM25 only

results

10

Search

1 · QUERY

Query understanding

normalise · extract identifiers · detect query type

47 ms

2 · RETRIEVAL CHANNELS

gte-modernbert-base

semantic · 768-d · 149M params · top 300

300 found · 481.6 ms

BM25

keywords · code-aware tokenizer · top 200

200 found · 33.8 ms

Qwen3-Embedding 0.6B

semantic · instruction-aware · top 300

300 found · 417.8 ms

Exact Identifiers

23,033 inexact · top 50

3 · CANDIDATES

Candidate union

up to 500 · every candidate scored by every channel

500 candidates

4 · EVIDENCE

34 ranking signals

5 families · similarity keywords, identifiers, I/O literals, agreement, length

5 · RANKING

Hybrid score fusion

gte + Qwen3-Embedding-0.6B cosines (second-encoder share 0.73) + BM25 + QLSR: the learned ranker is used for problem statements

short query · 1.8 ms

6 · RESULTS

Verified code

re-read from the content store by hash-calibrated confidence

answered from APPS P0 corpus

8,765 units searched

verified source

10 results

ranked by Hybrid score fusion

over 500 candidates from gte-modernbert-base, BM25, Qwen3-Embedding 0.6B

1049 ms in the engine

Technical details

1

class Solution totalQueens() dfs() d195

Semantic · gte-modernbert-base #1

Semantic · Qwen3-Embedding-0.6B #1

Keywords · BM25 #5

found by 3 of 3 channels

high confidence

class Solution:

def totalQueens(self, n):

def dfs(list, xy_dif, xy_sum):

p=len(list)

if p==n: res.append(list)

for q in range(n):

if (q not in list) and (p-q not in xy_dif) and (p+q not in xy_sum):

dfs(list+[q], xy_dif+[p-q], xy_sum+[p+q])

res=[]

dfs([],[],[])

return len(res)

cosine 8.7731 cosine (2nd encoder) 8.6186 BM25 7.81 content hash 5437780778016296 392 bytes

nQueen() d1796

Semantic · gte-modernbert-base #2

Semantic · Qwen3-Embedding-0.6B #2

found by 2 of 3 channels

def nQueen(n):

if n==2 or n==3: return []

r, odds, evens = n%6, list(range(1,n,2)), list(range(0,n,2))

if r==2:

evens[::2] = evens[::2][::-1]

evens.append(evens.pop())

if r==3:

odds.append(odds.pop())

evens.extend(evens[::2])

del evens[::2]

return odds+evens

cosine 8.7378 cosine (2nd encoder) 8.7467 content hash 83a3cc4048dbd089 316 bytes

check() passAll() reconstruct() class Solution +2 more d1892

Semantic · gte-modernbert-base #4

Semantic · Qwen3-Embedding-0.6B #3

Keywords · BM25 #15

found by 3 of 3 channels

from copy import deepcopy

from collections import deque

def check(p1, p2):

if p1[0] == p2[0] or p1[1] == p2[1] or abs(p1[0] - p2[0]) == abs(p1[1] - p2[1]):

ACIS

Hybrid code retrieval · learned ranking · offline

Search

APPS corpus

Versions

P1 · one pinned version

Evolution

Bonus · every version

Evaluation

P0 · recorded benchmark

Architecture

models · channels · ranker

1 · Content-addressed store

every unit stored once, by the hash of its bytes

2 · Immutable snapshot per version

a query sees exactly one version, never a mix

3 · Incremental build

only units whose content changed are embedded

4 · Atomic activation

rollback and crash recovery: no half-built index served

check if a string is a palindrome

repository · apps-history

version · latest (active)

or selector · snapshots...

as_of:2026-09-01

Search

answered from repository apps-history

version v15

390 units searched

verified source

10 results

ranked by Hybrid score fusion

over 344 candidates from gte-modernbert-base, BM25

81 ms in the engine

Technical details

1

palindrome() solution_8098.py

Semantic · gte-modernbert-base #1

Keywords · BM25 #5

found by 2 of 2 channels

high confidence

def palindrome(num):

if not isinstance(num, int) or num < 0:

return 'Not valid'

s = str(num)

return sum(s[i:len] == s[i:len][::-1] for i in range(len(s)-1)) for n in range(2, len(s)+1))

cosine 8.7145 BM25 5.72 content hash 9612a37f8117876 212 bytes version v15

2

class Solution breakPalindrome() solution_9835.py

Semantic · gte-modernbert-base #2

Keywords · BM25 #2

found by 2 of 2 channels

class Solution:

def breakPalindrome(self, palindrome: str) -> str:

if len(palindrome) % 2 == 1:

return palindrome[0:len(palindrome)-1]

else:

mid = len(palindrome) // 2

if palindrome[mid] != min(palindrome[mid], palindrome[mid+1]):

return palindrome[:mid] + min(palindrome[mid], palindrome[mid+1]) + palindrome[mid+1:]

return palindrome

Versions of apps-history

LABEL UNITS SNAPSHOT

v1 400 s_8ffc630359e

v2 400 s_36a0e945116

v3 398 s_37762a745e

v4 395 s_14250b02f4f

v5 390 s_4936c86337d

v6 390 s_336a56e583

v7 390 s_35a2f09642

v8 390 s_4c81ef86d7

v9 390 s_4c80fa23e5

v10 390 s_45fce2aa82

v11 390 s_346d3c247

v12 390 s_14302b2896

v13 390 s_4e411857ec

v14 390 s_9518b026d7

v15 390 s_4134873a88 active

Roll back active version

New commit

synthetic edits

3

class Solution removePalindromeSub() s08031.py

Semantic · gte-modernbert-base #3

Keywords · BM25 #7

found by 2 of 2 channels

class Solution:

def removePalindromeSub(self, s: str) -> int:

"a"

"aba"

"bab"

"abba"

"abba"

"abba"

"abba"

if len(s) == 0:

return 0

show all 15 lines

cosine 8.6788 BM25 5.53 content hash 8c29c49c2d81346 307 bytes version v15

4

is_isogram() solution_8097.py

Semantic · gte-modernbert-base #4

Keywords · BM25 #48

found by 2 of 2 channels

from collections import Counter

import re

def is_isogram(word):

if type(word) is not str or not word: return False

return len(set(Counter(re.sub(r"[^a-z]", "", word.lower())).values())) == 1

cosine 8.6279 BM25 6.88 content hash c4c40b03372ac47 201 bytes version v15

5

hamster_me() solution_0148.py

Semantic · gte-modernbert-base #5

Keywords · BM25 #106

found by 2 of 2 channels

def hamster_me(code, message):

code, dct = sorted(set(code)), {}

for c1,c2 in zip(code, code[1:] + [chr(ord("z") + ord(code[0]) - ord("a"))]):

for n in range(ord(c1), ord(c2)+1):

dct[chr(''-(n-97)*26 + 97)] = c1 + str(n-ord(c1)+1)

return ''.join(dct[c] for c in message)

cosine 8.6281 BM25 6.88 content hash f062956f1080478 301 bytes version v15

[illegible]

Every thing below is read from the running engine, the generated results and the badge what it looks, how large it is, and what it means.

1. QUERY

Query understanding

Knowledge about identifiers, select, every type

2. NETWORK CHANNELS

gen-modembert-base

channel: 716.2
100k queries, top 200

3. Query Embedding-0.8

channel: 400
Instruction points: top 200

3. CANDIDATES

Candidate scores

up to 500 every candidate scored every channel

4. EVIDENCE

3x ranking signals

8 features: similarity, keywords, identifiers, VC details, agreement, length

5. RANKING

learned ranking

LightGBM+LambertLink
400 Top, based on 5,000 queries

6. RESULTS

sorted cuts

are read from the channel every 10k, validated, confidence

Models

Alibaba-NLP/gen-modembert-base

PRIMARY 250K CHANNEL

channel	documents	length	weight
100M	70k	1024	20k MB

channel: 400
Instruction points: top 200

apex-2.0

channel: 400
Instruction points: top 200

Query/Query Embedding-0.8

SECOND 250K CHANNEL

channel	documents	length	weight
200M	1024	1024	1.5k MB

channel: 400
Instruction points: top 200

apex-2.0

channel: 400
Instruction points: top 200

Learned ranking

Rank: 400

Ranking: 1k

Ranking: 5,000 queries

Learned ranking

Rank: 400

Ranking: 1k

Ranking: 5,000 queries

Encoder selection

gen 0.7k, all 5,000 doc queries

ENCODER	PARAMS	MODQ/DOC	MODQ/DOC	DECISION	LOGGERS	
gen-modembert-base	selected	100M	71.2k	67.4k	2.6k	most accurate pre-trained cut point 2.6k, inside the 0.5 budget
embedding-english-v2		100M	56.7k	58.4k	2.7k	not good for the first 10k queries, relevance is 0.7k
gen-embedding-small-english-v2		40M	53.8k	50.2k	0.7k	17.8k predicted for the first 10k queries, relevance is 0.7k

Since retrieval above, each encoder on all 5,000 doc queries, on the CPU. Rule gen 0.7k, every pre-trained, because CPU requires models of at most 100 parameters, the smallest under 1400000 of all the best for relevance within 0.7k when the best models could not reach 0.7k. Decision: gen-modembert-base.

Learned and identifier channels

channel	documents	length	weight
100M	70k	1024	20k MB

channel: 400
Instruction points: top 200

apex-2.0

channel: 400
Instruction points: top 200

Burtime

channel	documents	length	weight
100M	70k	1024	20k MB

channel: 400
Instruction points: top 200

apex-2.0

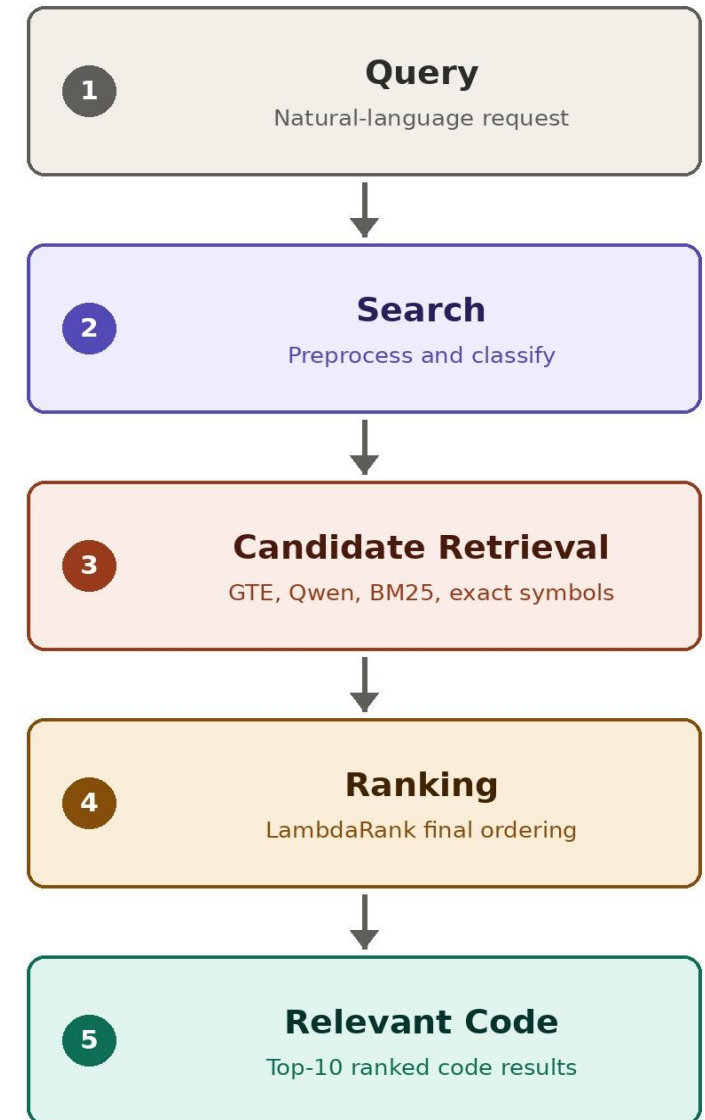
channel: 400
Instruction points: top 200

Demo & Product Walkthrough

- **1. Enter Query**
Developer enters a natural-language code query.
- **2. Understand & Search**
ACIS preprocesses the query and searches through semantic, lexical, and symbol indexes.
- **3. Retrieve Candidates**
Relevant code candidates are combined from **GTE, Qwen, BM25, and Exact Symbol Search**.
- **4. Rank Results**
LambdaRank ranks the candidates using 34 evidence signals.
- **5. Show Relevant Code**
The system displays the **Top-K code snippets with file paths and line locations**.

DEMO VIDEO (YOUTUBE LINK): <https://youtu.be/1m9tNrEVDPE>

ACIS Demo Flow



Results: Official TEST and DEV

Metric	Result
Official TEST: NDCG@10	78.283 (MTEB, CoIR AppsRetrieval test split, 3,765 queries)
Official TEST: MRR@10	74.393 · Recall@10 90.44 · Recall@100 98.33
DEV: NDCG@10 / MRR@10	87.07 / 84.72 (5,000 queries, out of fold)
DEV: gain over baselines	+16.04 NDCG@10 vs gte dense (71.03); +12.94 vs previous ACIS (74.13)
DEV: rank of the right code	#1 for 79.4% of queries · top 10 for 94.3% · in candidate pool for 99.4%
P1: version update	1 to 10 changed units searchable in about 2 s (p95)

Tools and tech stack used

Category	Technology
Language	Python 3.12
Package Management	uv
Embeddings	GTE-ModernBERT (768D) + Qwen3-Embedding-0.6B (1024D)
Lexical Search	BM25
Exact Code Search	Identifier / Symbol Search
Code Parsing	Python AST
Ranking	LightGBM LambdaRank (34 features)
Data Storage	SQLite + Content-Addressed Zstandard Blob Store
Vector Storage	Memory-Mapped .npy Vectors + Vector Cache
Backend API	FastAPI + Uvicorn
Validation / Models	Pydantic
Metrics & Logging	Prometheus-style Metrics + Structlog
Evaluation	MTEB 2.21.0 + pytrec_eval
Testing & Quality	Pytest + Hypothesis + Ruff + Mypy
Deployment	CPU-first / Offline

Impact & Use case

Impact	Use Case
Faster code discovery	Quickly find relevant code in large repositories
Higher developer productivity	Reduce manual code-search effort
Better AI coding support	Provide precise code context to AI agents
Efficient debugging	Locate related functions and implementations
Large-scale code handling	Search across thousands of files and functions
Version-aware development	Retrieve relevant code from specific versions

Innovation highlights, results and limitations

Innovation / Feature	Result / Status
Hybrid Retrieval	Combines two dense encoders, code-aware BM25 and exact identifier search
Contract-Aware Ranking	34 query/code evidence features, incl. encoder agreement, for learned ranking
Candidate Union	99.38% candidate recall (DEV) before final ranking
Intelligent Ranking	TEST: 78.283 NDCG@10, 74.393 MRR@10 · DEV: 87.07 NDCG@10, 84.72 MRR@10
Retrieval Accuracy	TEST Recall@10 90.44%, Recall@100 98.33% · DEV Hit@1 79.4%, Hit@10 94.3%
Dense-Only Improvement	+16.04 NDCG@10 over dense-only gte on DEV (71.03 to 87.07)
Evidence-Gated Code Lineage	Tracks modified, moved, renamed and replaced code while rejecting low-confidence links
Version-Aware Retrieval	Supports immutable snapshots, incremental updates and rollback
Explainable Retrieval	Provides confidence, ranking signals and retrieval timings
CPU-First Offline Architecture	No API keys or GPU required at query time
Qwen3-Embedding	Active: second dense encoder; DEV NDCG@10 74.13 to 86.70 when added
SPLADE / Cross-Encoder / LLM Reranker	Not used by design: no language model in the ranking path
Known limitations	TEST (78.28) is below DEV (87.07), nothing tuned on TEST; passes 6 of 9 robustness limits
Compute cost (CPU only)	Cold official run: 27,849 s (7.7 h) on 8 cores with both encoders; no GPU

What's next

- **Cut Encoding Time** with faster Qwen3 inference and lighter indexing
- **Harden Robustness** on the perturbation families that exceed their limits
- **Extend Language Support** beyond Python through AST-aware ingestion
- **Optimize Large-Scale Search** for faster indexing and retrieval
- **Add Agentic Query Refinement** on top of the retrieval core: iterative search and validation
- **Deploy as a Developer Tool** for real-world repositories and coding workflows

Brownie points slide(differentiation)

Feature	Differentiation
Hybrid Retrieval	Two dense encoders + code-aware BM25 + exact symbols, fused by a learned LambdaRank
Verified Retrieval	Every result is re-read from a content-addressed store by hash; nothing is generated
AST-Aware Intelligence	Understands code structure beyond plain text
Version-Aware Retrieval	Immutable snapshots, atomic activation and instant rollback
Code Evolution Tracking	Evidence-gated lineage: 323/323 lineages recovered, 0% wrong merges, duplicate rate@10 0.00 vs 0.77 flat
CPU-First Design	No API keys or GPU: the official 78.283 NDCG@10 was produced on an 8-core CPU

Checklist- Updated on Public GitHub

- Working prototype code — public or shared GitHub repo (Y/N)
- README with reproducible setup instructions (Y/N)
- Demo video, max 5 minutes (YouTube or Drive link)
- Presentation file (PPT or PDF) (Y/N)

Github Official Release:

https://github.com/abhinavakku513-collab/PRISM_GENAI_HACKATHON_Y2026/releases/tag/v1.0.0

**“The future of AI-assisted software engineering is not just about generating code.
It is about understanding the code that already exists.”**

From natural-language intent to verified source code - ACIS turns code retrieval into an intelligent, evidence-driven process.

Semantic understanding.

Multi-channel retrieval.

Learned ranking.

Version-aware intelligence.

ACIS - making large codebases intelligible to humans and AI agents.

Retrieve. Rank. Trace. Evolve.

Turning massive codebases into intelligent, navigable knowledge.

Thank You.

Team_JARVIS/ Samsung PRISM GenAI Hackathon - Theme 1